

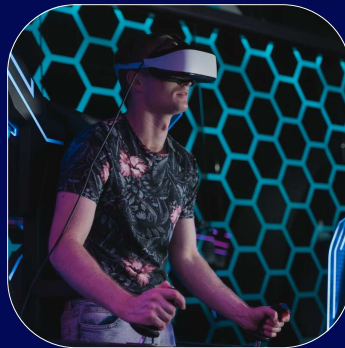
An abstract graphic consisting of multiple thin, light blue lines that form a series of overlapping, wavy shapes across the center of the image, resembling a stylized sound wave or a topographical map. The waves are more pronounced on the left and right sides, framing the central text.

OSCILLATION 3.0

IDEATHON

PITCH DECK

OUR TEAM



Tejas More

Team Member

PROBLEM STATEMENT

THE MODEL DEGRADATION CHALLENGE

AI models degrade silently in real-world environments

STATIC MODELS vs DYNAMIC DATA

Models are trained on historical data but deployed in constantly changing environments, leading to performance decay.

DATA DRIFT

User behavior, market trends, and external factors continuously shift data distributions beyond what models were trained on.

NO DECISION LAYER

Current systems detect issues but do not decide what action should be taken, leaving adaptation dependent on human intervention.

LIFECYCLE MANAGEMENT GAP

Lack of automated lifecycle management leads to delayed response and unreliable AI systems.

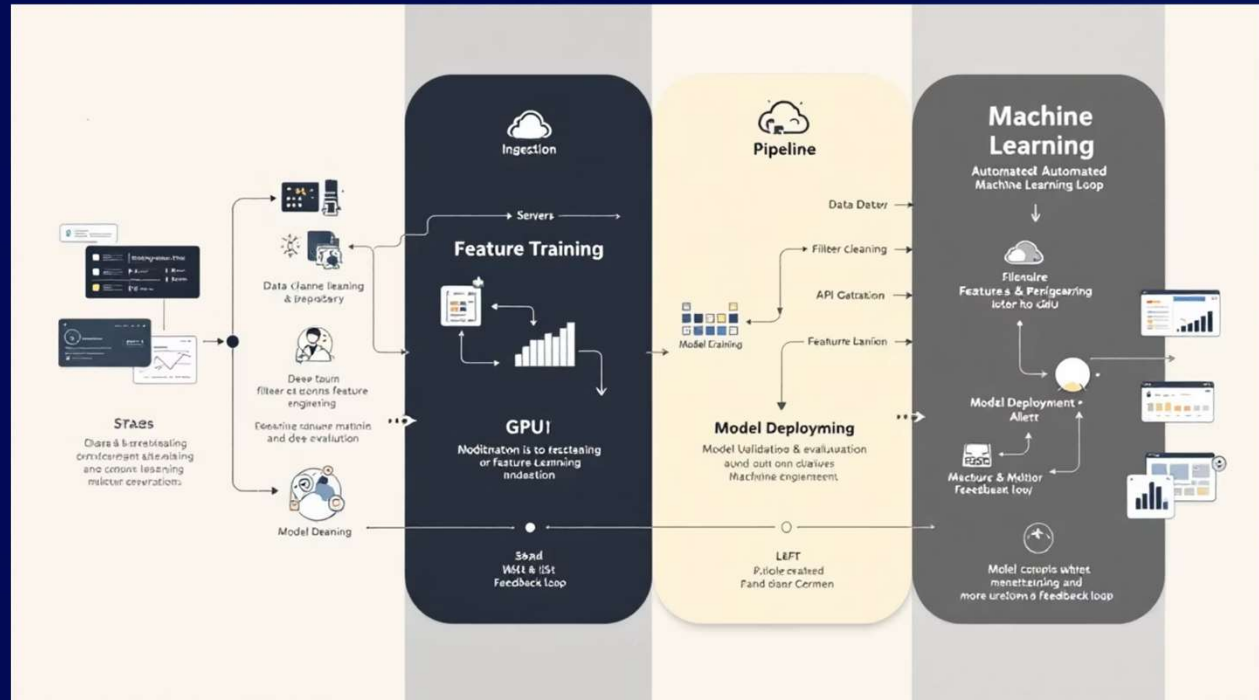


PROPOSED SOLUTION

AUTONOMOUS DECISION INTELLIGENCE SYSTEM

“Æther AI introduces a decision layer that determines when and how ML models should evolve.”

- Continuous monitoring of model performance and data behavior
- Multi-signal analysis (drift + performance + data quality)
- Decision engine that evaluates whether adaptation is required
- Conditional retraining (not periodic or manual)
- Model versioning and controlled deployment
- Identifies root cause of performance degradation
- Generates human-readable explanations
- Bridges technical outputs with business understanding



GAPS IN CURRENT ML SYSTEMS

Gap 1: Fragmented Ecosystem

Monitoring, retraining, and explainability exist as separate tools with no unified control mechanism.

Gap 2: Reactive Adaptation

Actions are triggered after performance drops, rather than proactively evaluating system health.

Gap 3: No Decision-Making Layer

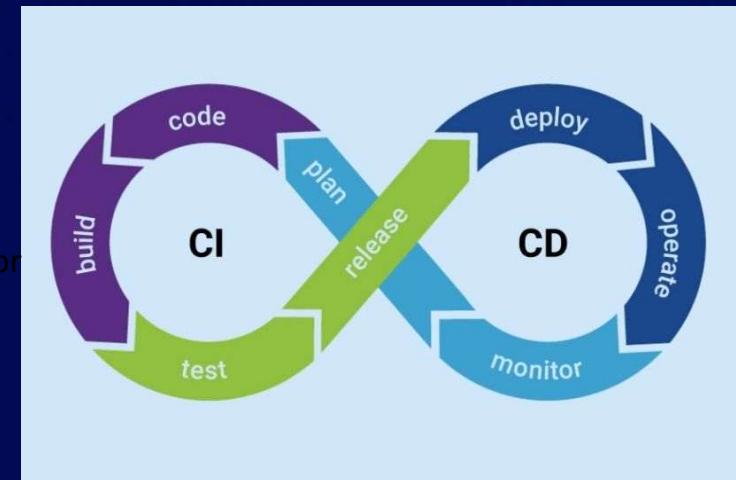
Systems detect anomalies but do not determine the optimal action, leaving critical decisions to humans.

Gap 4: Lack of Business Interpretability

Outputs are technical, making it difficult for non-technical stakeholders to understand and act on insights.

Gap 5: High Operational Dependency

Maintaining ML systems requires constant involvement of data teams, increasing cost and complexity.



NOVELTY

CORE INNOVATION: DECISION INTELLIGENCE FOR ML SYSTEMS

“We don’t automate ML pipelines – we enable them to decide when and how to evolve.”

01

CORE INNOVATION

- Introduces a system that determines when and how models should evolve
- Moves beyond monitoring → into decision-making
- Treats ML lifecycle as an adaptive system, not a static pipeline

03

TECHNICAL INNOVATION

- Multi-signal decision logic (not single-metric triggers)
- Feature-level drift interpretation for root cause analysis
- Explainable decision outputs (why action was taken)
- Modular architecture enabling scalable integration

02

KEY DIFFERENTIATORS

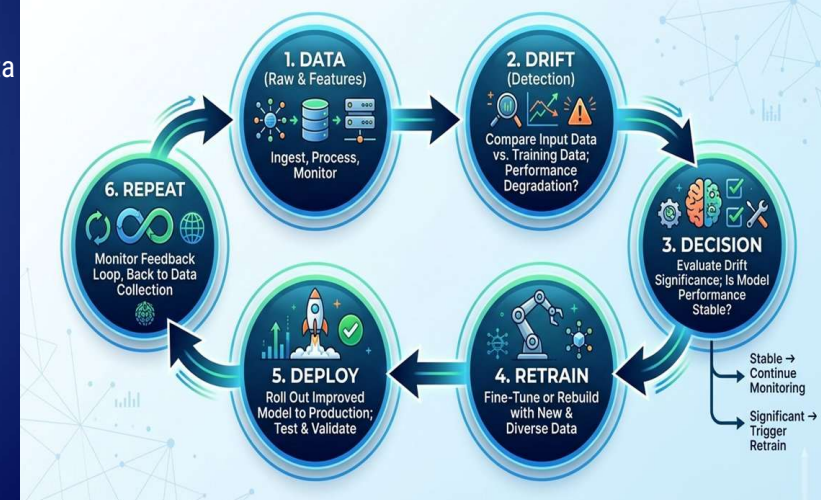
- Decision-based retraining (not time/schedule-based)
- Multi-signal evaluation (drift + performance + data quality)
- Conditional action system (adapt only when necessary)
- Built-in feedback loop for continuous improvement

04

WHY IT MATTERS ?

- Prevents unnecessary retraining → reduces cost
- Improves reliability of ML systems in dynamic environments
- Reduces dependency on continuous human monitoring
- Makes AI systems more trustworthy for business use

CLOSED-LOOP AI/ML MODEL LIFECYCLE DIAGRAM



MARKET & COMPETITIVE POSITIONING

01

Market Opportunity

- Rapid growth in AI adoption across industries
- Increasing reliance on ML models in real-time decision systems
- Rising need for continuous model reliability and performance stability
- Growing demand for tools that maintain ML systems post-deployment

02

Core Market Gap

- Monitoring tools detect issues but do not decide actions
- MLOps platforms manage pipelines but lack adaptive intelligence
- Explainability tools provide insights but no operational response

No system integrates detection + reasoning + action into a unified decision layer.

03

Key Competitors

- Evidently AI → Monitoring only (no action layer)
- Fiddler AI → Explainability-focused (no adaptation capability)
- WhyLabs → Observability and diagnostics (no decision system)
- AWS SageMaker / Azure ML → Pipeline orchestration (manual or scheduled retraining)

04

Our Positioning

- Decision intelligence layer on top of existing ML systems
- Works alongside current tools (not replacing infrastructure)
- Enables conditional, data-driven model adaptation
- Reduces unnecessary retraining and operational cost

We don't replace MLOps – we make it intelligent.

COMPARATIVE POSITIONING TABLE
MARKET LANDSCAPE

COMPETITOR TYPE	FEATURE FOCUS	DRIFT ACTION	COMPLEXITY	PRICING
 Evidently AI	Monitoring only	Alerts only	✓ Medium	Moderate
 Fiddler AI	Explainability & Ethics	Passive Analysis	✗ High	Enterprise-tier
 WhyLabs	Observability	Diagnostics	✓ Medium	Volume-based
 AWS SageMaker / Azure ML	Heavy Enterprise MLOps	Manual Retraining Trigger	✗ Very High	Usage-heavy
OUR CLOSED-LOOP SYSTEM	Unified Monitoring + Adaptation + Insights	Automated Decision & Retrain	✓ Low (SME-Friendly)	Value-based

REVENUE GENERATION

1 Primary Revenue Model

- SaaS-based platform with subscription tiers
- Pricing based on:
 - Number of models monitored
 - Data volume processed
 - Level of automation (monitoring → full decision system)

2 Who Pays & Why

- SMEs and data-driven startups using ML in production
- Companies facing model performance degradation over time
- Organizations lacking dedicated ML monitoring teams

They pay to reduce model failure risk and operational overhead

3 Pricing Strategy

- Prevents revenue loss due to poor model predictions
- Reduces unnecessary retraining costs
- Minimizes dependency on ML engineers
- Ensures consistent model performance in dynamic environments

4 Revenue Expansion

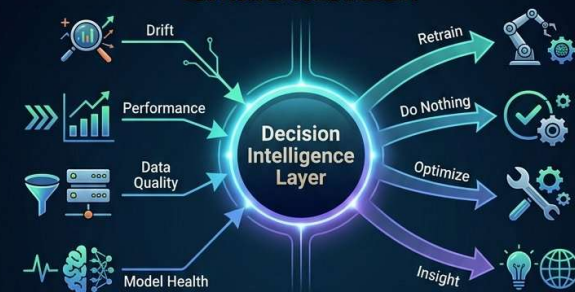
- Enterprise integrations and custom deployments
- API-based pricing for large-scale usage
- Advanced analytics and insight modules
- Consulting and optimization services

Revenue Model & Value Flow

SECTION 1: VALUE CREATION



SECTION 2: CORE INSIGHT



SECTION 3: BUSINESS VALUE



SECTION 4: REVENUE MODEL



SECTION 5: EXPANSION



The image features a dark blue background with a central graphic of glowing blue lines. These lines form a large circle with several smaller arcs and dots, creating a sense of motion or a network. On the left and right sides of the central circle, there are faint, wavy lines that resemble sound waves or data signals. The text "THANK YOU" is prominently displayed in the center in a bold, white, sans-serif font.

THANK YOU